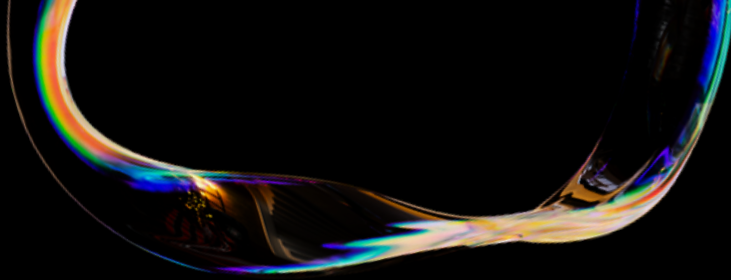




21 DAYS SQL CHALLENGE

DAY 21

BY AMRUHA AHMED



DAILY CHALLENGE

Create a comprehensive hospital performance dashboard using CTEs.

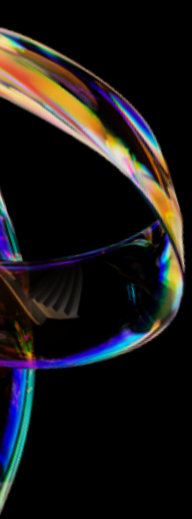
Calculate:

- 
- 1) Service-level metrics (total admissions, refusals, avg satisfaction),
 - 2) Staff metrics per service (total staff, avg weeks present),
 - 3) Patient demographics per service (avg age, count).

Then combine all three CTEs to create a final report showing service name, all calculated metrics, and an overall performance score (weighted average of admission rate and satisfaction). Order by performance score descending.

AMRUHA AHMED

DAILY CHALLENGE



WITH service_metrics AS (
SELECT
service,
SUM(patients_admitted) AS Total_Admissions,
SUM(patients_refused) AS Total_Refusals,
AVG(patient_satisfaction) AS Average_Satisfaction
FROM services_weekly
GROUP BY service)

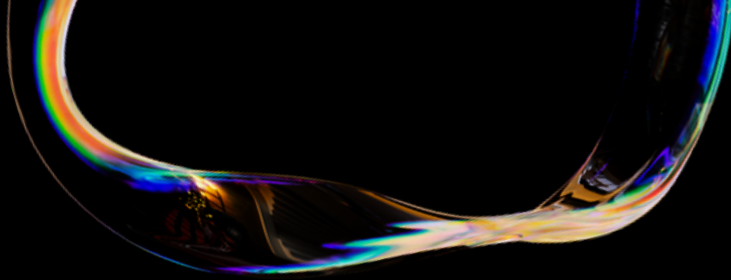
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DAILY CHALLENGE

```
,staff_metrics AS  
(  
  SELECT  
    service,  
    COUNT(DISTINCT staff_id) AS Total_Staff,  
    ROUND(AVG(weekly_present)) AS Average_Weeks_Present  
  FROM  
    (  
      SELECT service, staff_id, COUNT(*) AS weekly_present  
      FROM staff_schedule  
      WHERE present=1  
      GROUP BY service,staff_id  
    )d GROUP BY service)
```

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DAILY CHALLENGE



```
,patient_metrics AS  
(  
  SELECT  
    service,  
    ROUND(AVG(age),0) AS Average_Age,  
    COUNT(DISTINCT patient_id) AS Patient_Count  
  FROM patients  
  GROUP BY service)
```

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DAILY CHALLENGE

```
SELECT
service_m.service,
service_m.Total_Admissions,
service_m.Total_Refusals,
service_m.Average_Satisfaction,
staff_m.Total_Staff,
staff_m.Average_Weeks_Present,
patient_m.Average_Age,
patient_m.Patient_Count,
((service_m.Total_Admissions/(service_m.Total_Admissions+service_m.Total_Refusals)) *0.3 +
service_m.Average_Satisfaction*0.7) AS Performance_Score
FROM service_metrics service_m
JOIN staff_metrics staff_m
ON service_m.service=staff_m.service
JOIN patient_metrics patient_m
ON staff_m.service=patient_m.service
ORDER BY performance_score DESC;
```

AMRUHA AHMED

DAILY CHALLENGE



service	Total_Admissions	Total_Refusals	Average_Satisfaction	Total_Staff	Average_Weeks_Present	Average_Age	Patient_Count	Performance_Score
ICU	648	141	81.6154	34	31	44	241	57.37717
general_medicine	2332	1938	81.2308	28	31	46	242	57.02540
surgery	1686	555	79.2692	25	31	46	254	55.71414
emergency	1185	5008	77.8846	39	31	45	263	54.57662



AMRUHA AHMED

The background features two large, flowing, and swirling liquid-like shapes on a solid black background. These shapes are highly reflective and iridescent, displaying a spectrum of colors including red, orange, yellow, green, blue, and purple. They appear to be made of a viscous, glossy material, possibly oil or paint, that has been manipulated to create these fluid, organic forms. The central text is positioned between these two decorative elements.

THANK YOU

for your time and attention

AMRUHA AHMED